Veda Technology Business & Service Analytics

Major Project Report — Data Science Track

Objective

This project analyzes simulated business data to identify website traffic patterns, internship and training demand, IT service inquiries, customer segments, revenue trends and opportunities for data-driven decisions.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook, Excel, Power BI, SQL and Git/GitHub.

Dataset

A synthetic dataset containing 500 records was created for educational analysis. It includes marketing channel, service, customer type, website visitors, internship applications, training enrollments, IT inquiries, service requests and revenue.

Analysis

The workflow includes data quality checks, exploratory analysis, monthly trend analysis, service-demand analysis, customer segmentation, correlation analysis and KPI reporting.

Predictive Modeling

A Random Forest regression model is included to estimate revenue from business activity indicators. MAE and R^2 are used for evaluation.

Recommendations

Monitor high-demand services, improve acquisition channels using visitor data, plan training capacity from enrollment trends, use customer segmentation for targeted communication, and maintain a recurring KPI dashboard.

Conclusion

The project demonstrates an end-to-end introductory data science workflow from preprocessing through analysis, visualization, prediction and business recommendations. All data is synthetic.